STAFF REVIEW · H446 005 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q5

Do Now — source answers

1. 01101000 0011 6.5 = 110.1_2 = $0.1101_2 \times 2^3$. Mantissa 01101000 ; exponent 0011 .

2. 01101101_2 and 109 . 6 → 0110 , D → 1101 . In denary: (6 × 16) + 13 = 109 .

3. −10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

4. $00000110_2 = 06_{16}$ AND with 00001111 clears the upper nibble and preserves the lower nibble 0110 .

I Do — source worked steps

1. Evaluate the mantissa — CODE:
01011000
= ½ + ⅛ + 1⁄16
= 0.6875 — Read the mantissa as a signed binary fraction.

2. Convert the exponent to denary — CODE:
0011 = 3 — The exponent is a signed two’s-complement integer.

3. Apply the scale — CODE:
0.6875×2^3
= 0.6875 × 8
= 5.5

We Do — guided CHECK

1. Build the answer

2. Build the answer

3. Build the answer

4. Build the answer
Shift the mantissa left twice and reduce the exponent by 2.

Page 3 — independent skill check, teacher-review model

1. 01110000 = +0.875, exponent 0010 = +2, so 3.5. 00011010 is not normalised: its first two mantissa bits are 00.

You Do — source mark schemes

1. CODE:
Mantissa 0100 = 0.5
Exponent 1110 = -2

0.5 × 2^-2 = 0.125

Answer = 0.125

2. CODE:
Mantissa 1010 = -0.75
Exponent 1110 = -2

-0.75 × 2^-2 = -0.1875

Answer = -0.1875

3. CODE:
a) 010101 100 → normalised
b) 110101 111 → not normalised
c) 011010 010 → normalised
d) 101010 110 → normalised

Rule:
positive normalised mantissa begins 01
negative normalised mantissa begins 10

4. CODE:
(a) 01110000 0010

mantissa = 0.1110000_2 = 0.875
exponent = 2

0.875 × 2^2 = 3.5

(b) 10010000 1111

mantissa = -1 + 1/8 = -0.875
exponent = -1

-0.875 × 2^-1 = -0.4375

5. CODE:
More mantissa bits:
→ greater precision / more significant binary digits.

More exponent bits:
→ greater range / larger and smaller magnitudes can be represented.

If the total number of bits is fixed:
allocating more bits to the mantissa improves precision but leaves fewer bits for the exponent and reduces range;
allocating more bits to the exponent increases range but leaves fewer bits for the mantissa and reduces precision.

Exit Ticket — source answer

1. False. An exponent changes the magnitude; the sign of the mantissa determines whether the value is positive or negative.